

Protocols — Guided Notes

Unit 1.5.2 | CompTIA Network+ N10-009 Obj. 1.5 | N10-008 1.5

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

ICMP does not carry _____ data. It operates at Layer _____ and does not use _____. The two most common uses of ICMP in troubleshooting are _____ and _____.

Question 2

If ping fails to a host that is actually up and running, the most likely reason is _____. This means a failed ping does NOT automatically mean _____.

Question 3

ARP resolves _____ addresses to _____ addresses. A device broadcasts an ARP _____ to the segment, and the target responds with a unicast ARP _____. The result is stored in the device's _____ table.

Question 4

Write out the DHCP DORA process:

D — _____

O — _____

R — _____

A — _____

Question 5

A rogue DHCP server occurs when _____. The symptom on affected workstations is _____. To find the rogue server, you would _____.

Question 6

In the DNS resolution process, if the local DNS server doesn't have a cached answer, it queries a _____ name server first, which directs it to the _____ server for the domain's extension, which directs it to the _____ name server that has the actual IP address.

Question 7

Before HTTP data moves over HTTPS, the client and server perform a _____ handshake. The server presents a _____ to prove its identity. All traffic after this point is _____ on port _____.

Question 8 — Real World Application

REAL WORLD: A user reports 'the internet is down.' You run ping 8.8.8.8 — it succeeds. You run ping google.com — it fails. Walk through your diagnosis and fix.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1-5-2-protocols/1-5-2-protocols-answer-key.pdf
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